بسم الله

TECH SOVEREIGNTY: BUILDING WHAT WE CONTROL

Binyamin Barsch

UMMAH TECH

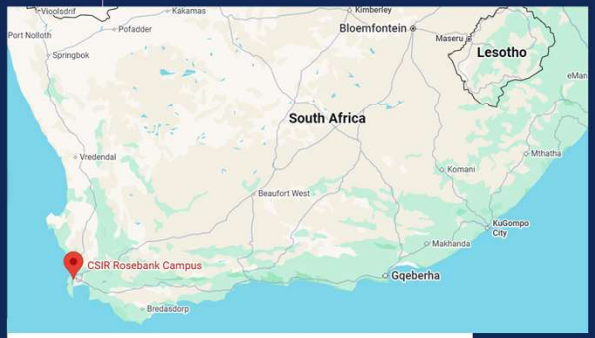
FAITHMEETSTECHNOLOGY

CENTRE FOR HIGH PERFORMANCE COMPUTING

NATIONAL SUPERCOMPUTING FACILITY

- 1PF system
- Formerly Top500
- Largest HPC System in Africa (YMMV)





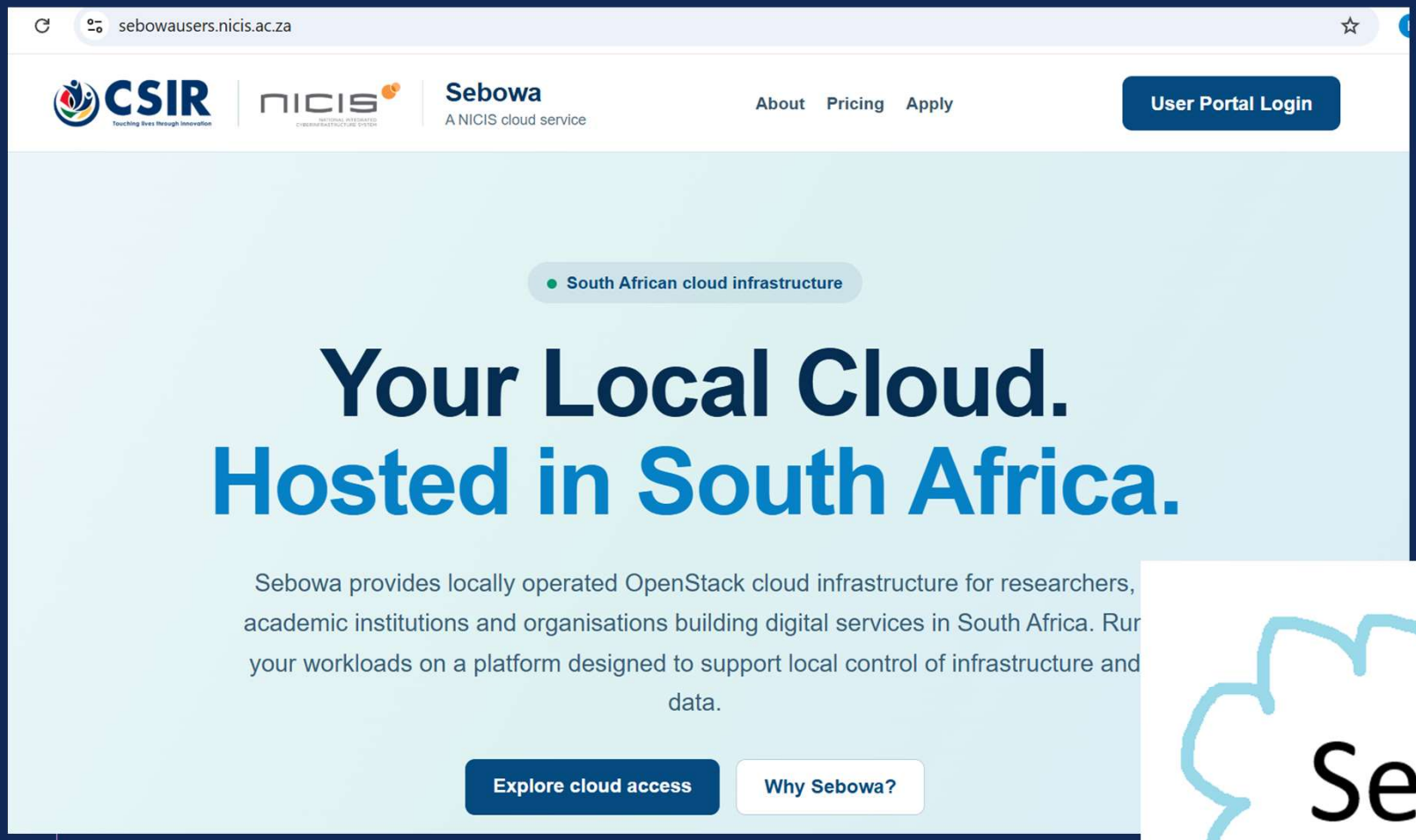
... FOR RESEARCH & INDUSTRY

- South African Research Institutions & Government
- Southern Africa Community partners
- Industry representatives



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Hosted in South Africa.
Sebowa provides locally operated OpenStack cloud infrastructure for researchers, academic institutions and organisations building digital services in South Africa. Run your workloads on a platform designed to support local control of infrastructure and data.
Explore cloud access
Why Sebowa?


Sebowa

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TRUST & SOVEREIGNTY

Sovereignty begins with the freedom to make our own decisions.

- Trust: Being given the freedom to make decisions and act independently.
- Autonomy: The ability to make choices without external interference.
- Ownership: Having control over our decisions, resources and direction.
- Responsibility: Being accountable for the choices we make.

Sovereignty is not just about independence, it is about having the power and responsibility to shape our own future.

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SOVEREIGNTY IN OUR COMMUNITIES

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South African Muslim communities have built, owned their needs.

- **Education:** Independent schools and Islamic educational institutions
- **Masjids:** Community-built and managed places of worship
- **Halaal Certification:** Community-established standards
- **Zakaat & Welfare:** Independent charitable and social support organisations
- **Finance & Business:** Community enterprises and Islamic financial services
- **Freedom of Expression:** Community media, public lectures and workshops
- **Moon Sighting:** Community-led decisions, celebration of Eid together!

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DO WE HAVE TECH SOVEREIGNTY?

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HISTORY OF DIGITAL BUSINESS

- 1990s–2000s: Businesses built and hosted their own digital infrastructure.
- 2010s: Digital infrastructure became increasingly centralised around major cloud and platform providers.
- 2020s: AI accelerated this concentration, but open models and more capable local hardware also created opportunities to run AI independently.
- 2030s: Possible move towards more open, interoperable and community-driven digital ecosystems.

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WHAT DO WE MEAN BY TECHNOLOGY?

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Technology is more than computers and software. It underpins everyday lives.

- **Businesses:** Digital enterprises, innovation and entrepreneurship
- **Skills & Employment:** Digital skills, software dev and job creation
- **Tools & Applications:** Microsoft Office, ERP, CRM, business software
- **Digital Services:** Websites, FinTech, e-commerce, AI platforms
- **Infrastructure:** Cloud computing, data centres, networks and cybersecurity

Technology sovereignty means having the ability to build, own, operate and make independent choices about the technology we depend on.

WHY NOW? WHAT HAS CHANGED? بسم الله

Technology has long been dominated by major corporations, but three developments are changing the landscape.

1. Geopolitics & Technology Dependency

- Essential digital tools, services and infrastructure concentrated among a few global corporations (Microsoft, AWS, Google, Oracle, OpenAI, Anthropic)
- Access influenced by geopolitical tensions, sanctions and corporate policies
- Risks of account suspensions, service restrictions, technology used in conflicts
- Ethical concerns: corporate involvement in wars, military contracts and BDS campaigns

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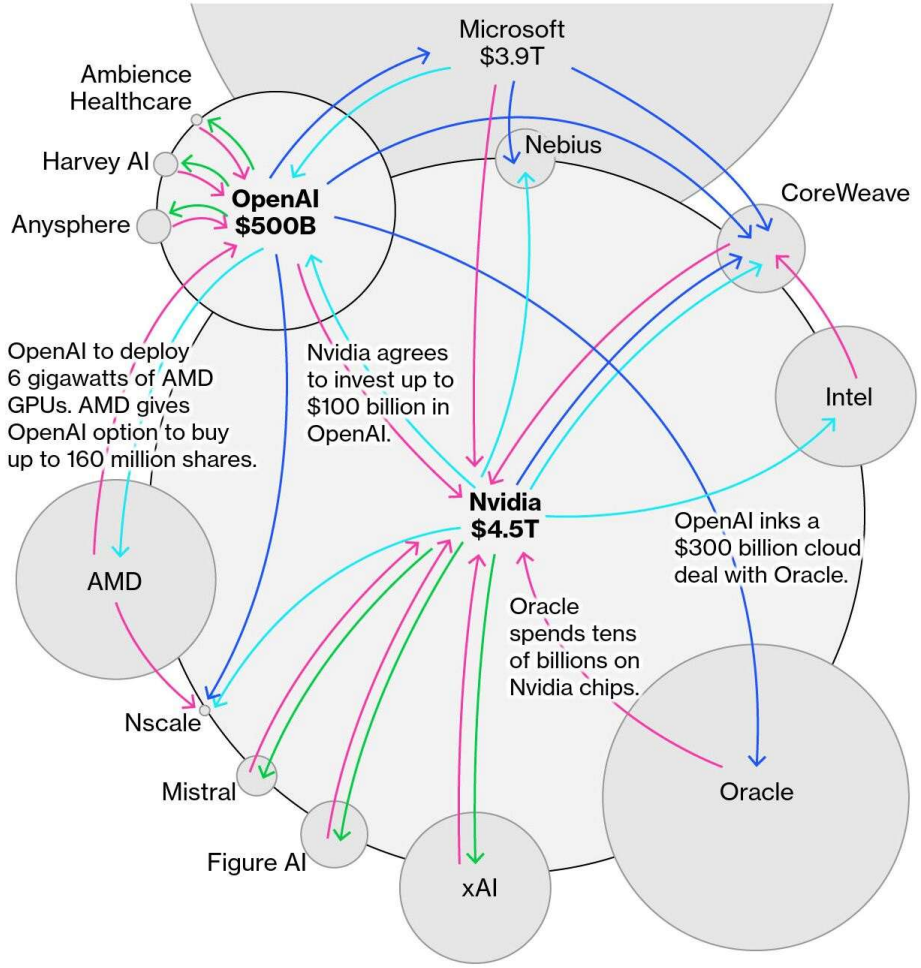
2. AI Revolution

- AI chatbots, tools and agents are boosting productivity, innovation and automation
- Growing reliance on AI risks weakening critical thinking, problem-solving and essential skills
- Increasing dependency on a few major AI providers…
- …Nvidia-Oracle-OpenAI/Anthropic

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How Nvidia and OpenAI Fuel the AI Money Machine

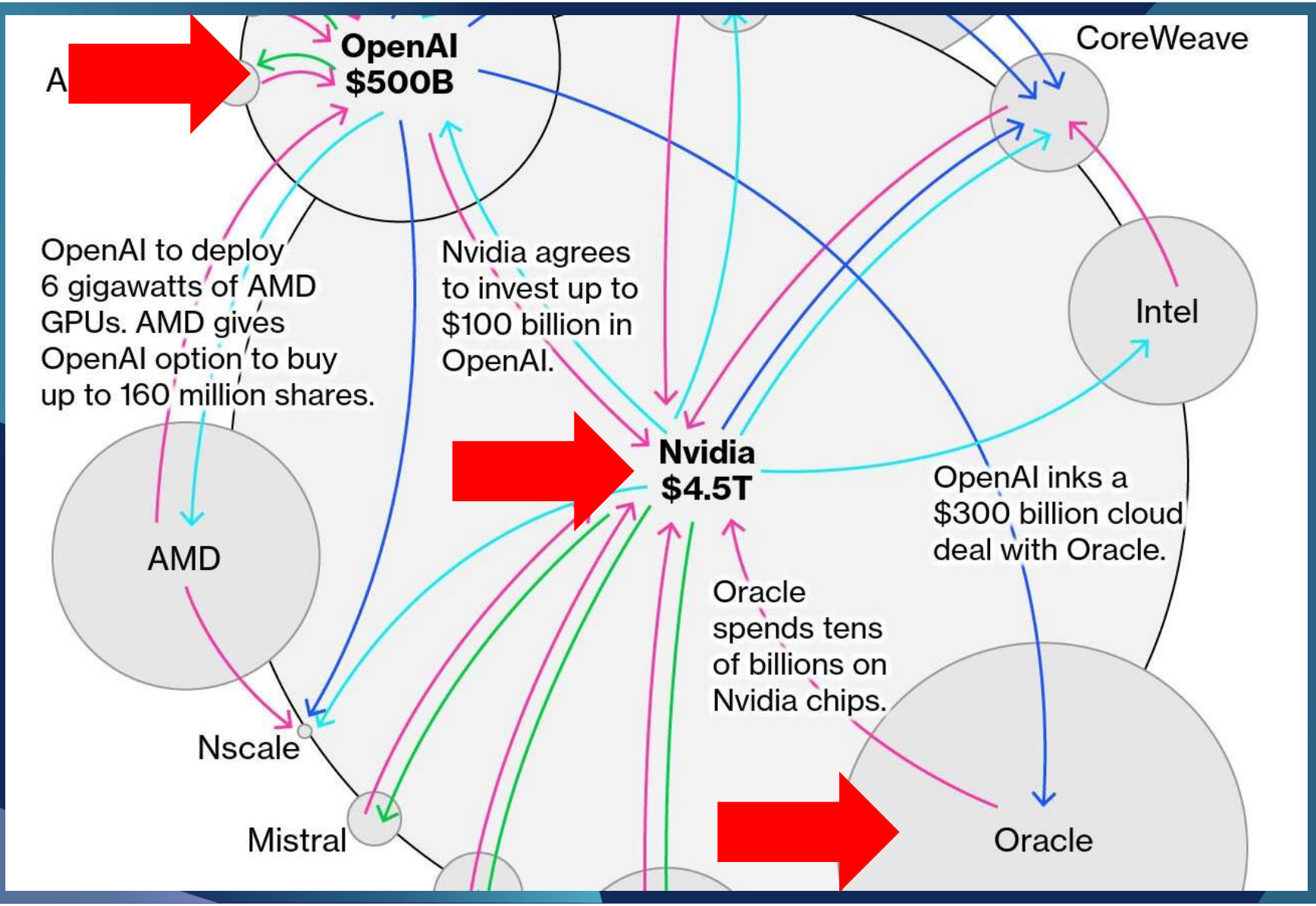
Hardware or Software · Investment · Services · Venture Capital
Circles sized by market value



Source: Bloomberg News reporting

Bloomberg

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A
OpenAI
$500B
CoreWeave
OpenAI to deploy
6 gigawatts of AMD
GPUs. AMD gives
OpenAI option to buy
up to 160 million shares.
Nvidia agrees
to invest up to
$100 billion in
OpenAI.
Intel
Nvidia
$4.5T
OpenAI inks a
$300 billion cloud
deal with Oracle.
AMD
Oracle
spends tens
of billions on
Nvidia chips.
Nscale
Mistral
Oracle

WHY NOW? WHAT HAS CHANGED? بسم الله

3. Cybersecurity & Data Sovereignty

- Increasing cyber threats, data breaches and privacy risks
- Loss of control over where data is stored, processed and accessed

The question is no longer just what technology we use, but who controls it.

Can you run your business without these tools and services?

If you can't doesn't that effect your sovereignty, your ability to choose?

HOW TO MAINTAIN TECH SOVEREIGNTY

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1. Invest in Local Infrastructure

- Develop local cloud, data centres and digital services
- Reduce dependency on foreign infrastructure providers

2. Adopt Open & Independent Alternatives

- Explore alternatives to Gmail, OneDrive, Microsoft, Oracle and cloud services
- Invest in open-source tools and community-driven platforms

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HOW TO MAINTAIN TECH SOVEREIGNTY

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3. Invest in People & Skills

- Develop local talent to build, operate and maintain technology
- Preserve critical thinking, problem-solving and technical skills
- Can we replace or maintain a tool if access is lost tomorrow?

4. Embrace Local & Offline AI

- Run open AI models locally, without relying entirely on external providers
- Maintain control over sensitive data, AI tools and infrastructure

HOW TO MAINTAIN TECH SOVEREIGNTY

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- Technology sovereignty is not just about owning technology, it is about having the skills, freedom and capability to control our digital future.]
- Do you think your subscription free protects you and your data?

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~~BUILDING~~ EXPLORING ALTERNATIVES

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EXPLORING ALTERNATIVES

Moving from dependency towards choice and control.

- **Open Source:** Explore community-driven alternatives to proprietary software
- **Everyday Tools:** Email, document editing, cloud storage, forms and websites
- **Business Applications:** ERP, CRM, finance and administration systems
- **Infrastructure:** Local cloud, self-hosting and on-premises solutions
- How many of our schools, madrassas and community organisations depend on Google Forms, Gmail and Microsoft 365?

We don't always need to build from scratch, we just need to choose differently.

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AI CHANGES WHAT IS POSSIBLE

AI is changing our ability to build and maintain technology.

- **Build:** Develop websites, applications and digital services faster
- **Customise:** Adapt open-source tools to community needs
- **Automate:** Reduce repetitive administrative and business tasks
- **Learn:** Use AI to develop technical skills and solve problems
- **Operate Locally:** Run open AI models on local computers and infrastructure

Small teams can make big changes, supported by AI and should help us become technology creators, not just technology consumers.

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INFRASTRUCTURE SOVEREIGNTY

Hosting locally does not automatically mean controlling locally.

- **Local vs Foreign Control:** An AWS data centre in South Africa is locally hosted, but remains operated by a global provider
- **Local Infrastructure:** Invest in South African-owned and operated cloud and data centre services
- **Data Sovereignty:** Control where sensitive data is stored, processed and accessed

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INFRASTRUCTURE SOVEREIGNTY

Hosting locally does not automatically mean controlling locally.

- **Critical Sectors:** Education, healthcare, legal services, finance and government
- **Digital Resilience:** Maintain the ability to operate independently when external services become unavailable

True sovereignty is not just where our data lives, it is who controls the infrastructure, access and decisions.

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TIE YOUR CAMEL.

Explore, Prepare & Have a Backup Plan

- **Explore Alternatives:** Know what other tools, services and providers are available
- **Avoid Single Points of Failure:** Don't depend entirely on one company, platform or AI provider
- **Have a Contingency Plan:** What happens if your email, cloud, website or AI service goes down tomorrow?

بسم الله

TIE YOUR CAMEL.

Explore, Prepare & Have a Backup Plan

- **Maintain Essential Skills:** Can your business still operate without AI agents and automated workflows?
- **Backup & Recovery:** Keep control of your data, backups and the ability to migrate or restore services

Technology should strengthen our organisations, not become the reason they collapse when something goes wrong.

TECH ALTERNATIVES

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Tool / Service	Alternatives	Self-Hosted
Microsoft Office (Word, Excel, PowerPoint)	LibreOffice, ONLYOFFICE	LibreOffice, Collabora
Google Drive / OneDrive	Proton Drive, Infomaniak kDrive	Nextcloud
Gmail / Outlook	Proton Mail, Tuta	Mailcow
Google Docs	CryptPad, ONLYOFFICE	Nextcloud + ONLYOFFICE
Google Forms / Microsoft Forms	Tally, LimeSurvey	Nextcloud Forms, LimeSurvey
Google Colab / Azure Notebooks	JupyterLab, JupyterHub	JupyterHub + JupyterLab
Microsoft Teams / Slack	Element, Wire	Mattermost, Matrix
AWS / Azure / GCP (VMs)	Hetzner, OVHcloud, Domains.co.za (SA)	OpenStack, Proxmox
AWS S3 / Cloud Storage	Scaleway, OVHcloud	Ceph, MinIO
ChatGPT / Microsoft Copilot	Mistral Le Chat	OpenWebUI + Ollama
GitHub Copilot / AI Coding Assistants	Mellum 2.1 (JetBrains)	Mellum 2.1 + llama.cpp
Shopify	Shopware, PrestaShop	WooCommerce, PrestaShop
Salesforce / Dynamics CRM	Odoo, SuperOffice	Odoo, ERPNext
Mailchimp	Brevo	Mautic, Listmonk
DocuSign / Adobe Sign	Yousign, Skribble	Documenso
Cloudflare / DNS	Bunny.net, deSEC	PowerDNS, Nginx
Datadog / Cloud Monitoring	European Grafana hosting	Prometheus, Grafana
Notion / Airtable	Anytype, Baserow	AppFlowy, Baserow

How a Website Can Be Hosted on a Raspberry Pi

Your Device
(laptop, phone, tablet)

1
You type the website address in a browser

Internet

2
Your request reaches your Raspberry Pi

Raspberry Pi
(in your home or local network)

Web Server
(e.g. Nginx or Apache)

Website Files
(HTML, images, etc.)

3
The website is sent back to your device and displayed in your browser

1 You request the website

You enter the website address (e.g. https://mypi.local or a public domain) in a browser on your laptop, phone or tablet.

https://mypi.local

2 The request reaches your Raspberry Pi

The request comes through the internet (or your local network) to your Raspberry Pi, which is running a web server such as Nginx or Apache.



3 Your website is displayed

The web server on the Raspberry Pi serves the website files (HTML, images, etc.) back to your device, and the website appears in your browser.

https://mypi.local

TECH ALTERNATIVES

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Tool / Service	Alternatives	Self-Hosted
Google Colab	JupyterLab	JupyterHub
GitHub Copilot / AI Coding Assistants	Mellum 2.1 (JetBrains)	Mellum 2.1 + llama.cpp

JupyterLab + Mellum 2.1

JUPYTER — BUILT BY THE COMMUNITY, USED BY THE WORLD

Proof that open-source communities can build world-class technology

- **100% Open Source:** Free, community-governed and independently developed
- **Global Adoption:** Used in education, research, data science and AI
- **Community Ownership:** Developed collaboratively rather than a corporation
- **Economic Value:** Powers significant commercial activity without being a proprietary product ~ worth about $6 Billion
- **Local Deployment:** JupyterLab can run on our own infrastructure

You don't need to be Big Tech to build technology with global impact.

MELLUM 2.1 — AI ON OUR OWN TERMS

Powerful AI doesn't always require expensive cloud infrastructure.

- **12 Billion Parameters:** Only 2.5 billion active per token
- **Open Source:** Apache 2.0 licensed model from JetBrains
- **Local AI:** Can run on CPU and RAM using quantised models
- **AI Agents:** Designed for coding, reasoning, tool use and automated workflows
- **Privacy & Control:** Run AI locally without sending sensitive data to external providers

From renting AI intelligence to running AI on infrastructure we control.

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TOWARDS MEDINA OS

TOWARDS MEDINA OS

The building blocks already exist.

- **Open-Source Software:** Alternatives to everyday digital tools and services
- **Local Infrastructure:** Local cloud providers and self-hosted solutions
- **AI Technology:** Open models and tools that make development more accessible
- **People & Skills:** Local developers, businesses and communities

The opportunity is to bring these together into an open, community-driven digital ecosystem.

TOWARDS MEDINA OS

A Community-Driven Digital Ecosystem

An initiative to explore, develop and support technology solutions that empower our communities.

- **Open:** Built around open-source technologies and collaboration
- **Independent:** Encouraging local ownership and control
- **Accessible:** Making digital tools and services easier to adopt
- **Sustainable:** Developing local skills to operate and maintain solutions

From being consumers of technology to becoming creators and owners.

SUMMARY

- We already have a history of establishing and managing our own institutions.
- We have the technology, the tools and the opportunity to do the same in the digital world.
- We need to explore how we can bring these capabilities together, not necessarily to replace every existing one.
- This will give us the ability to choose, build and control the technology that matters to us.



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LINK TO SLIDES...SHUKRAN

Binyamin Barsch
bbarsch@gmail.com

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